

A2G200N1700ME3H

1700V SiC MOSFET Module



Technical Features

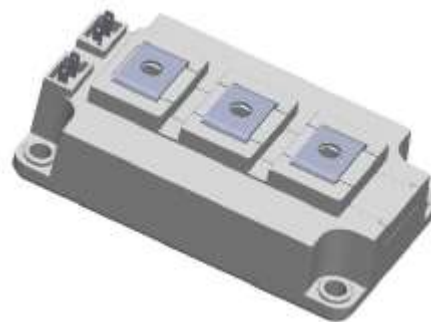
- Industry Standard 62 mm Footprint
- Ultra Low Loss, High-Frequency Operation
- Zero Reverse Recovery from Diodes
- Zero Turn-off Tail Current from MOSFET
- Normally-off, Fail-safe Device Operation
- Copper Baseplate and Aluminum Nitride Insulator

Applications

- High Frequency Switching application
- DC/DC converter
- Solar inverters and Wind Inverters
- UPS and SMPS
- Traction

Product Summary

V _{DS}	1700V
I _D	200A



Absolute Maximum Ratings (T_C=25°C unless otherwise specified)

Symbol	Parameter	Value	Unit
V _{DS}	Drain-Source Voltage	1700	V
V _{GS}	Gate-Source Voltage (dynamic)	-10/+22	V
I _D	Drain Current (continuous) V _{GS} =20V	200	A
I _{DM}	Drain Current (pulsed)	400	A
T _C , T _{stg}	Operating and Storage Temperature Range	-40 to +150	°C
T _J	Junction Temperature	175	°C
R _{θJC}	Thermal Resistance, Junction-to-heatsink	0.12	°C/W

Electrical Characteristics (T_C=25°C unless otherwise specified)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
BV _{DS}	Drain-source Break down Voltage	V _{GS} =0V	1700			V
I _{DSS}	Zero Gate Voltage Drain Current	V _{DS} =1700V, V _{GS} =0V			600	uA
I _{GSS}	Gate-body Leakage Current	V _{DS} =0V, V _{GS} =-10 to 20V			6	uA
V _{GS(th)}	Gate Threshold Voltage	V _{DS} =V _{GS} , I _D =60mA	2.0		4.0	V
R _{DS(on)}	Static Drain-source On Resistance	V _{GS} =18V, I _D =120A		12	14.7	mΩ
V _{GS(on)}	Recommended turn-on Voltage	Static		18		V
V _{GS(off)}	Recommended turn-off Voltage			-5		V
R _G	Gate Resistance	V _{GS} =0V, f=1MHz		0.5		Ω
C _{iss}	Input Capacitance	V _{DS} =1000V, f=1MHz V _{AC} =25mV		9.6		nF
C _{oss}	Output Capacitance			0.83		
C _{rss}	Reverse Transfer Capacitance			120		pF
E _{on}	Turn-On Switching Energy	V _{DS} =1000V, V _{GS} =-5/20V, I _D =150A, Load=150uH		3.8		mJ
E _{off}	Turn-Off Switching Energy			1.2		mJ
Q _{GS}	Gate-Source Charge	V _{DD} =1000V V _{GS} =-5/20V I _D =150A		270		nC
Q _{GD}	Gate-Drain Charge			132		
Q _G	Total Gate Charge			600		
t _{d(on)}	Turn-on Delay Time	V _{DD} =1000V V _{GS} =-5/20V, I _D =150A Load=150uH, T _J =175°C R _{ext} =4.7Ω		35		nS
t _r	Rise Time			22		
t _{d(off)}	Turn-off Delay Time			19		
t _f	Fall Time			15		

A2G200N1700ME3H

Typical Performance-Reverse Diode ($T_J=25^{\circ}\text{C}$ unless otherwise specified)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
V_{FSD}	Forward Voltage	$V_{GS}=0V$, $I_F=120A$		1.52	1.9	V
I_S	Continuous Diode Forward Current	$V_{GS}=0V$, $T_C=25^{\circ}\text{C}$		100		A
t_{rr}	Reverse Recovery Time	$V_{GS}=-5V$, $I_F=120A$ $V_R=1000V$ $di/dt=1000A/\mu s$, $T_J=175^{\circ}\text{C}$		36		ns
Q_{rr}	Reverse Recovery Charge			960		nC
I_{rrm}	Peak Reverse Recovery Current			63		A

Module Physical Characteristics

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
L_{Stray}	Stray Inductance			20		nH
W	Weight			340		g
M_s	Mounting Torque	M6-1.0 bolts	4		5.5	Nm
V_{isol}	Case Isolation Voltage	DC 1min	5.8			kV
	Clearance Distance	Terminal to Terminal		9		mm
		Terminal to Baseplate		30		mm
	Creepage Distance	Terminal to Terminal		30		mm
		Terminal to Baseplate		40		mm

A2G200N1700ME3H

Typical Performance

Figure 1. Output Characteristics ($T_J = 25^\circ\text{C}$)

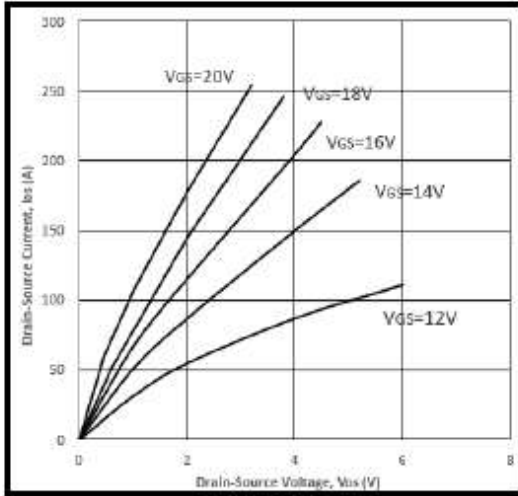


Fig 2. Normalized On-Resistance vs. T_J

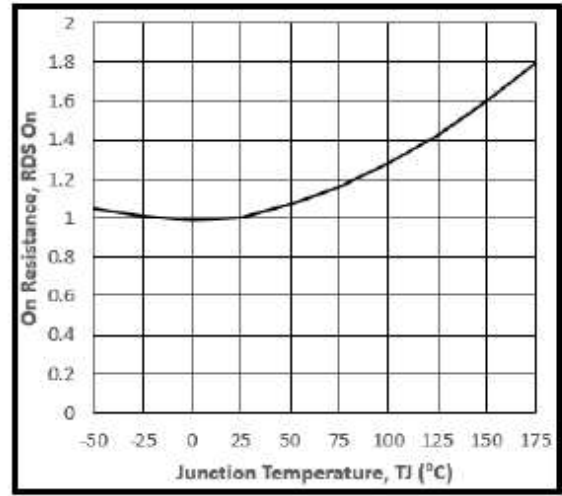


Figure 3. Threshold Voltage vs. Temperature

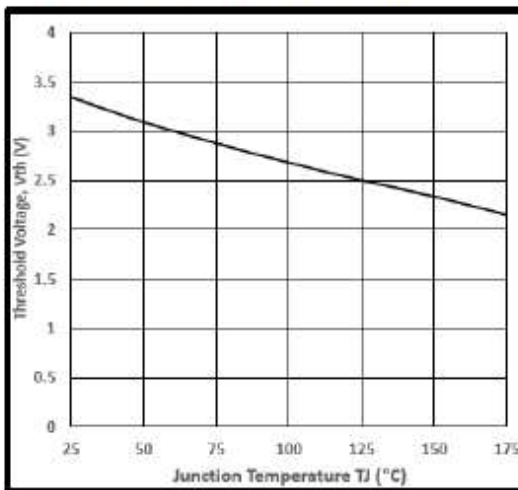


Figure 4. Transfer Characteristic for Various T_J

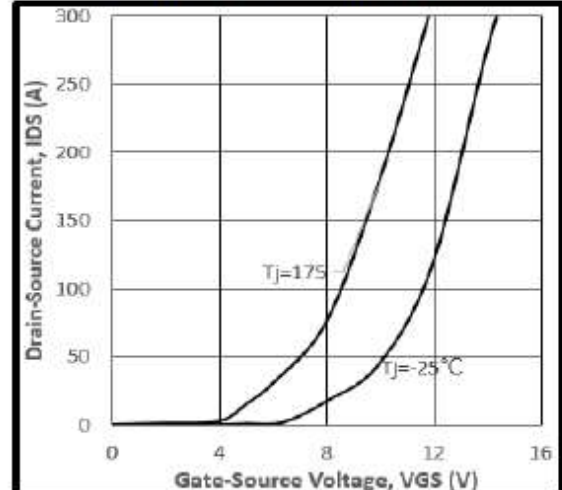


Figure 5. Diode Characteristic

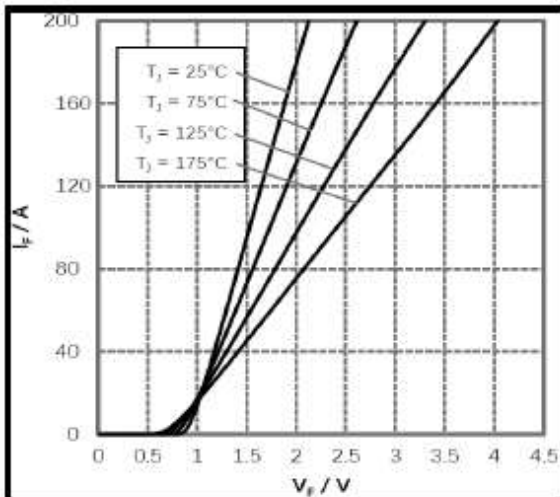
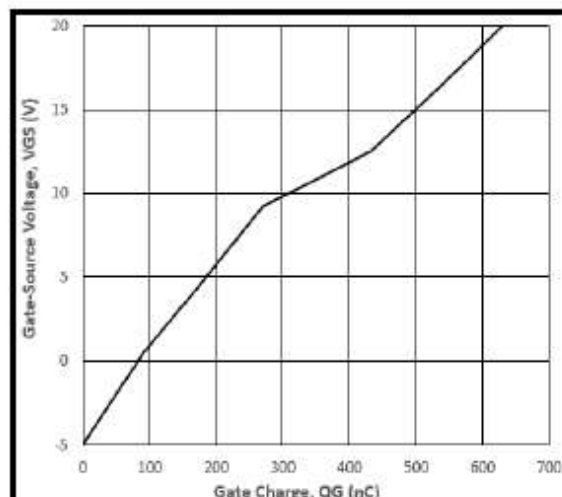


Figure 6. Gate Charge Characteristics



A2G200N1700ME3H

Figure 7. Capacitances vs. VDS

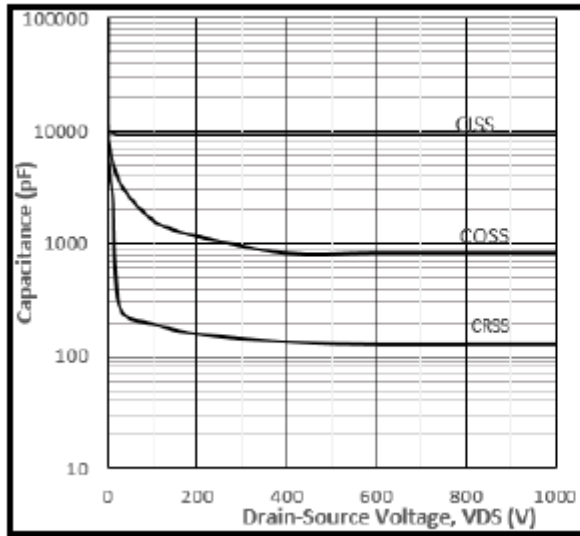


Figure 8. Inductive Switching Energy vs. Drain Current

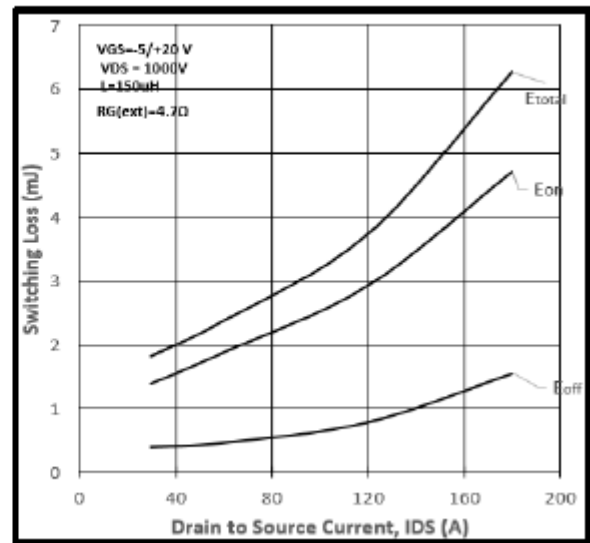


Figure 9. Switching Time Description

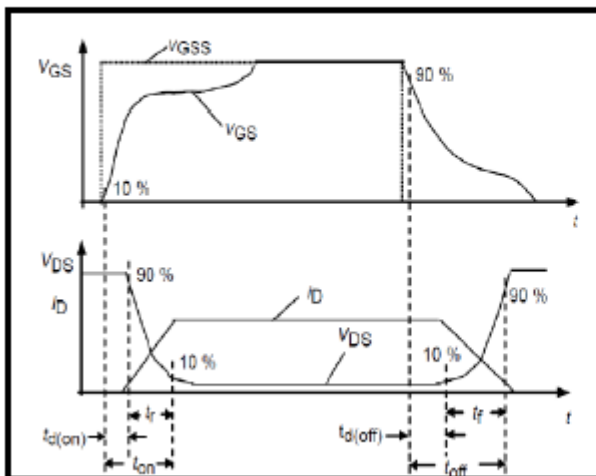
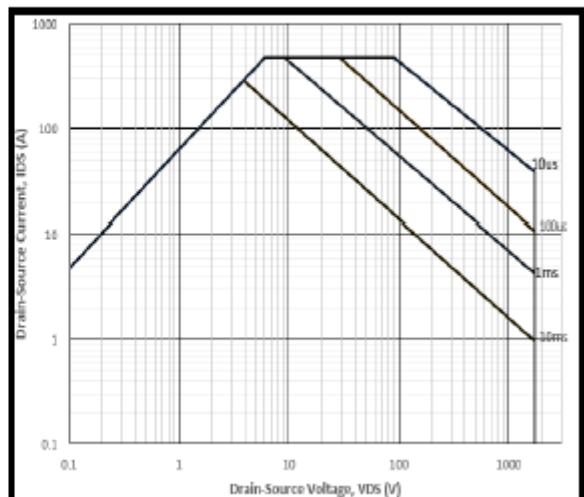
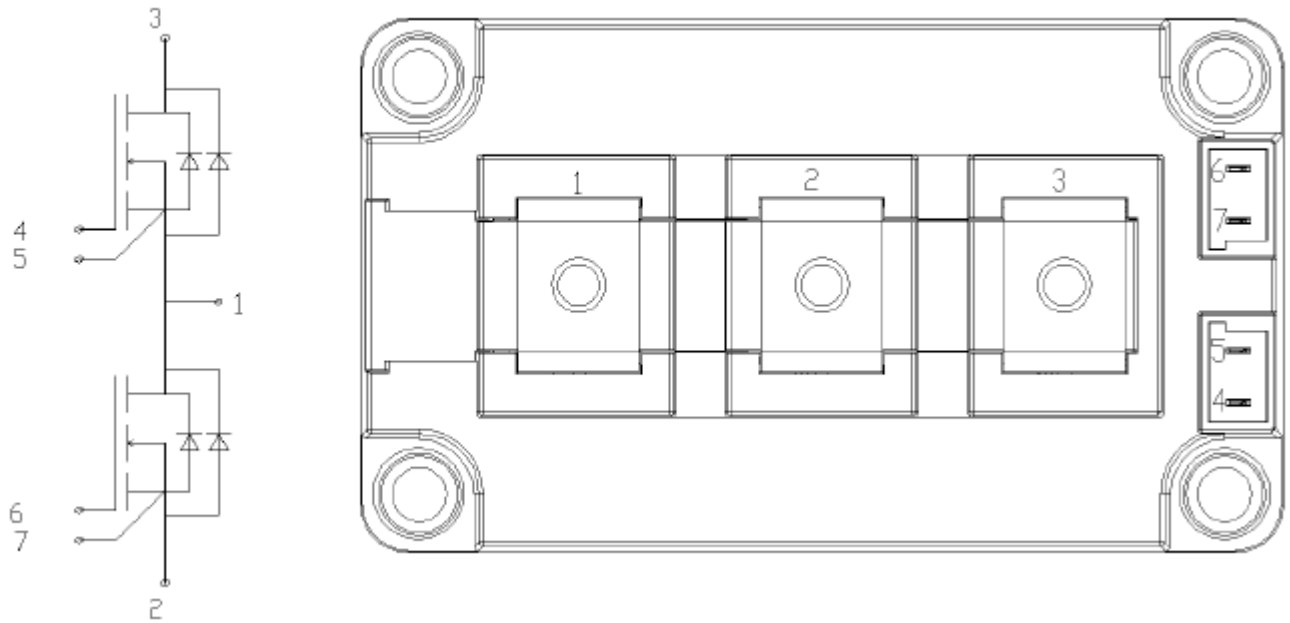


Figure 10. Safe Operating Area

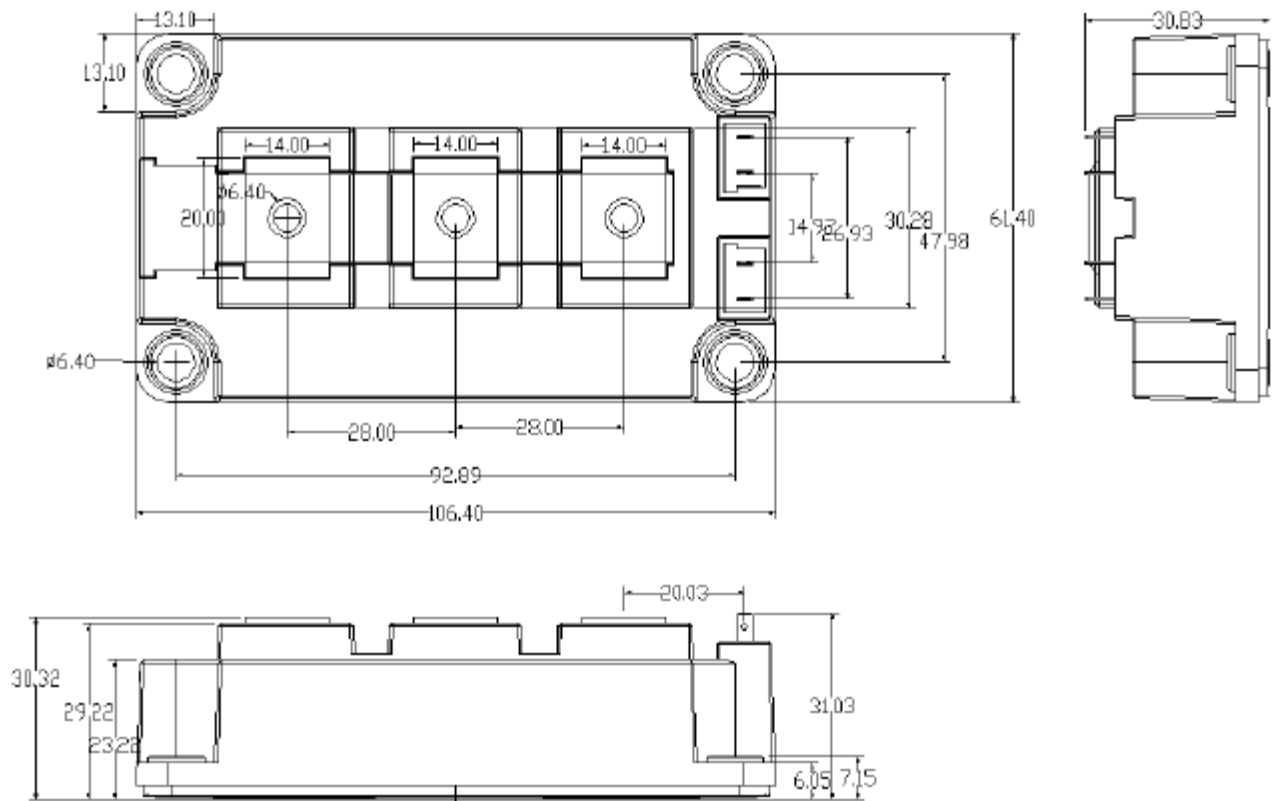


A2G200N1700ME3H

Circuit Diagram Headline



Package Drawing (mm)



Revision version	Description	Date
1	Initial	02.2024